Data Dominance: Your First Flight with Power BI

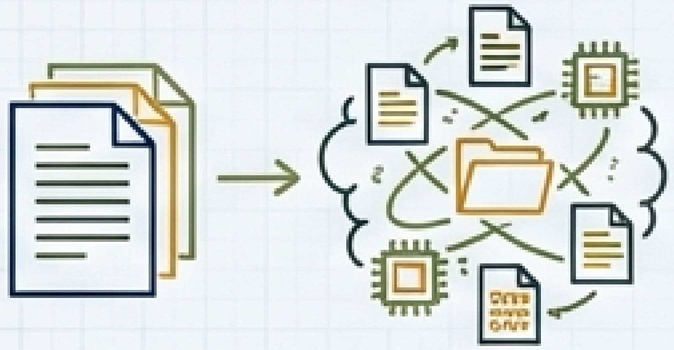
A **Foundational Briefing** for the California Air National Guard

Transforming Data into Decisive Action.

The Mission: From Data Overload to a Common Operational Picture

THE CHALLENGE – INFORMATION FOG

Our units generate vast amounts of data daily—from personnel records and equipment logs to mission readiness metrics. This data often lives in disparate systems, spreadsheets, and databases.



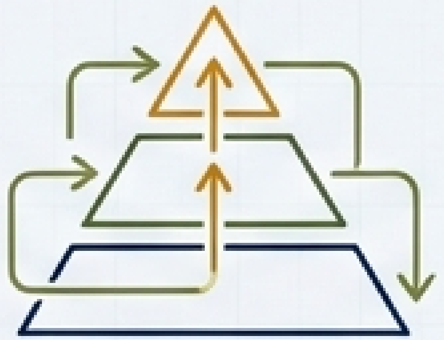
Slow, manual reporting processes.

Difficulty in getting a single, up-to-date view of readiness.

Valuable intelligence is siloed, hindering proactive decision-making.

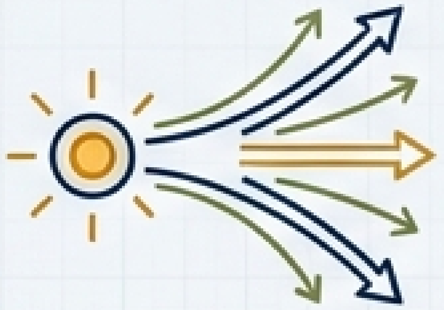
THE OBJECTIVE – ACHIEVE DATA DOMINANCE

Our goal is to leverage a strategic capability that provides a unified, interactive, and near real-time view of our most critical operational data.



Empower every level of command with actionable insights.

Accelerate the OODA Loop (Observe, Orient, Decide, Act).



Transform data into a true force multiplier.

Your Equipment: The Power BI Intelligence Platform

Power BI is a collection of software services, apps, and connectors that turn unrelated sources of data into coherent, visually immersive, and interactive insights.

The Power BI Ecosystem

Power BI Desktop
(The Hangar)

PUBLISH

Power BI Service
(The Air Operations Center)

ACCESS

Power BI Mobile
(In-the-Field Access)

Power BI Desktop
(The Hangar)

Power BI Desktop: The development environment. Where you connect, transform, model, and build reports on your local machine. This is the “hangar” where you build and prep your aircraft.

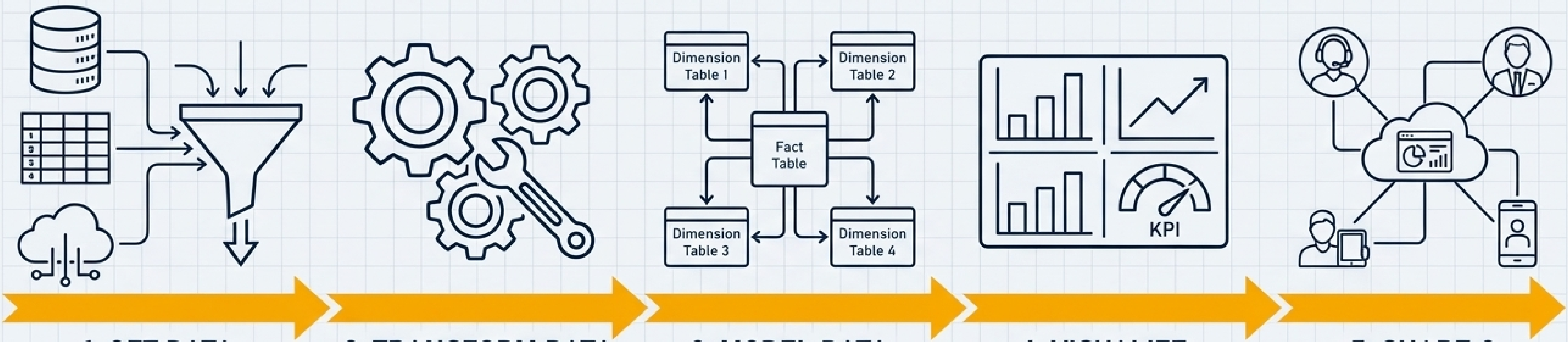
Power BI Service
(The Air Operations Center)

Power BI Service: The cloud-based collaboration hub (app.powerbi.com). Where you publish, share, and manage reports and dashboards. This is your “Air Operations Center”.

Power BI Mobile
(In-the-Field Access)

Power BI Mobile Apps: Access your intelligence on the go, with alerts to stay informed of critical changes. Your “in-the-field” data access.

The Operational Framework: From Raw Data to Actionable Intel



1. GET DATA

Connect to hundreds of data sources, from simple Excel files to complex SQL databases.

2. TRANSFORM DATA (Power Query)

Clean, shape, and standardize the raw data. This is where you ensure data quality *before* analysis.

3. MODEL DATA

Define relationships between your data tables. This step builds the logical foundation for accurate analysis.

4. VISUALIZE

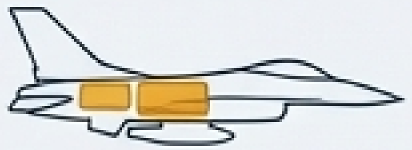
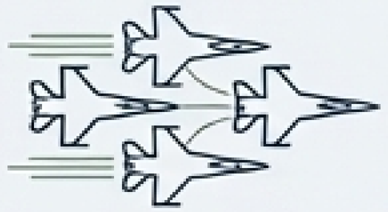
Create charts, graphs, maps, and KPIs to tell the story of your data in a report.

5. SHARE & COLLABORATE

Publish reports to the Power BI Service to create dashboards, share with stakeholders, and set up automated refreshes.

Pre-Flight Check 1: Your Refueling Strategy (Data Connectivity Modes)

Choosing how to connect to your data is the first critical decision. It defines the trade-off between report speed and data immediacy.

Connectivity Mode	Analogy: Refueling Strategy	Data Location	Performance	Data Freshness	Best For
Import	 Internal Fuel Tanks	A compressed copy is stored *inside* Power BI.	Very Fast (In-memory engine).	Stale until the next scheduled refresh (e.g., 8x/day for Pro).	Maximum performance and complex data modeling when real-time data is not required.
DirectQuery	 Live Mid-Flight Refueling	Data stays at the source. Power BI queries it live.	Depends entirely on the source database speed.	Real-Time.	Massive datasets that are too large to import, or when live data is a mission-critical requirement.
Live Connection	 Linking to an Existing Fleet's Systems	Connects to a pre-built enterprise model (Analysis Services or Power BI semantic model).	Depends on the linked model's performance.	Real-Time.	Ensuring a single version of the truth across the organization by leveraging a centrally managed data model.

This slide directly addresses a fundamental architectural choice. The analogies are key to making this technical concept intuitive for a new user.

Pre-Flight Check 2: The Aircraft Blueprint (Data Modeling & Star Schema)

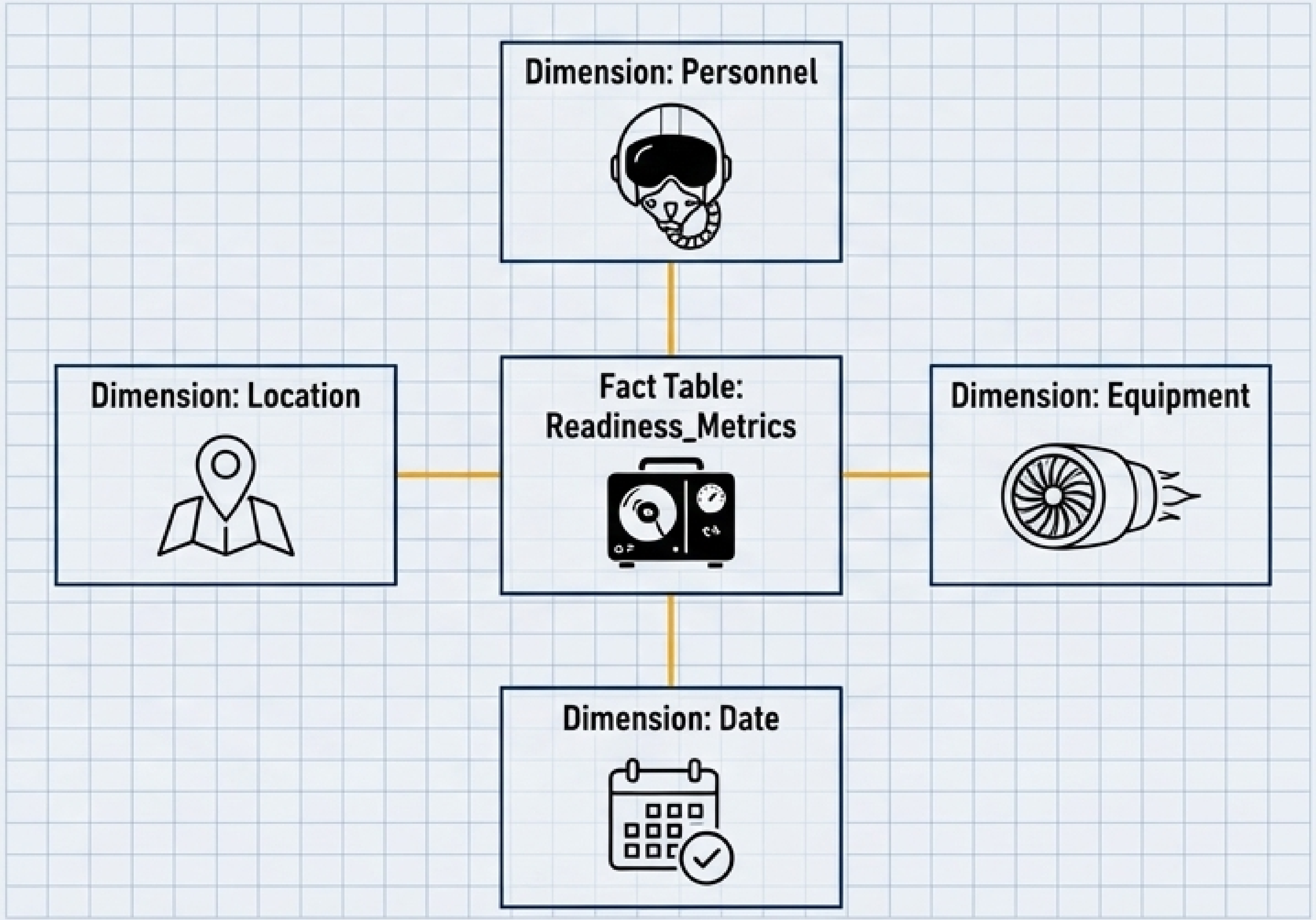
A well-designed data model is the foundation of a high-performance report. The universally preferred approach for Power BI is the **Star Schema**.

Analogy: The Blueprint for an Aircraft

Think of your data model as the engineering blueprint. A well-structured design ensures every component works together efficiently and reliably.

The Star Schema Explained

- **Dimension Tables (The Components):** Describe your business entities—the "who, what, where, when." (e.g., \`Personnel', \`Equipment', \`Date', \`Location'). They are the "wings, engine, fuselage."
- **Fact Tables (The Performance Data):** Store observations or events—the numbers and metrics. (e.g., \`Sorties Flown', \`Maintenance Hours', \`Mission Capable Rates'). They are the "flight hours, fuel consumed, actions taken."



Why It Matters: Power BI's engine is optimized for this structure. A star schema leads to smaller model sizes, faster queries, and a more intuitive experience for report users.

Pre-Flight Check 3: The Maintenance Crew (Power Query Editor)

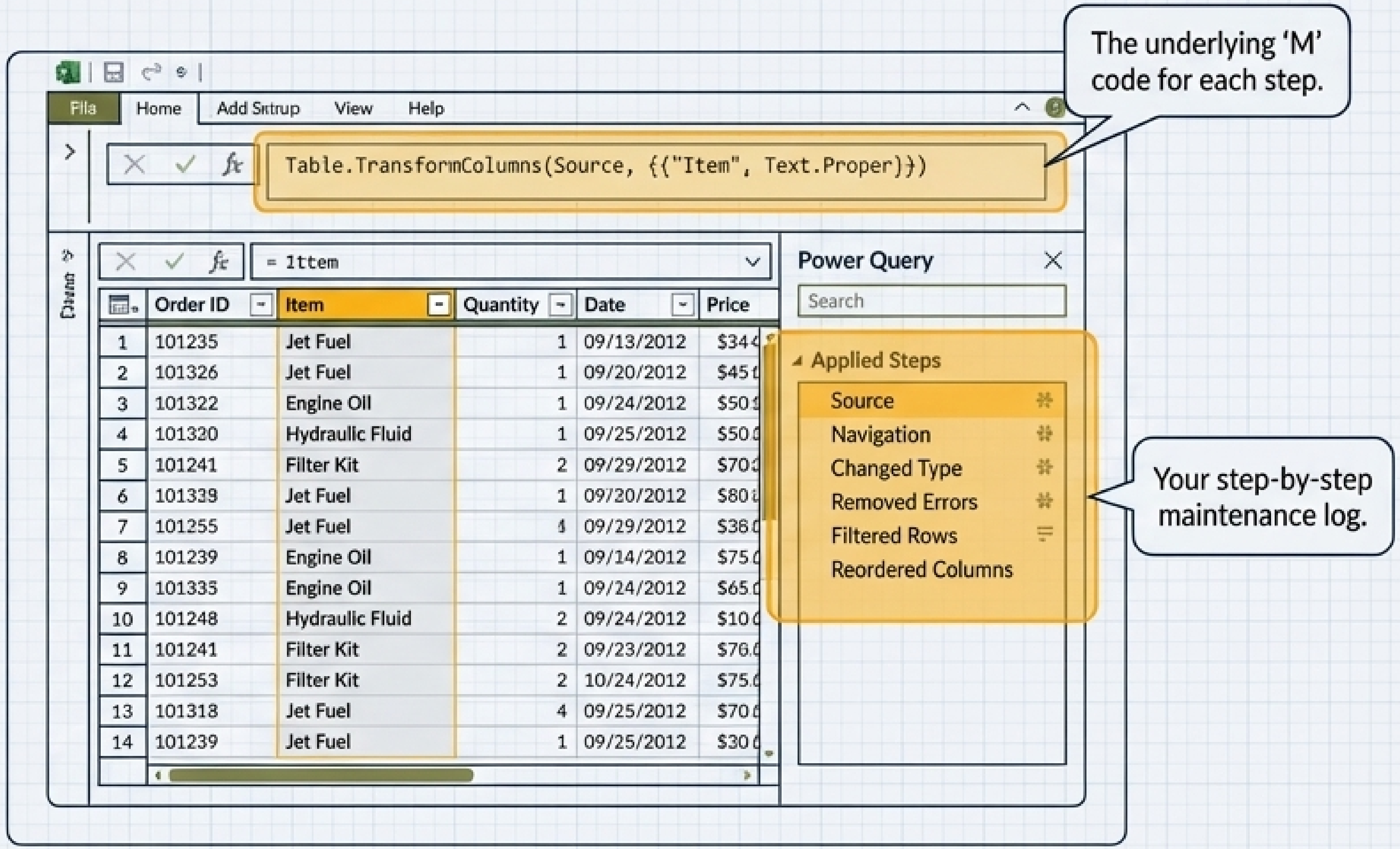
Power Query is your data maintenance crew. It sources, cleans, reshapes, and standardizes your "fuel" (data) *before** it's loaded into the "engine" (the data model), ensuring optimal performance and reliability.

Key Functions of Power Query

- Connect to hundreds of data sources.
- Remove errors, duplicates, and unnecessary columns.
- Change data types (e.g., text to number).
- Merge or append data from multiple tables.
- Create new custom columns.

The 'M' Language

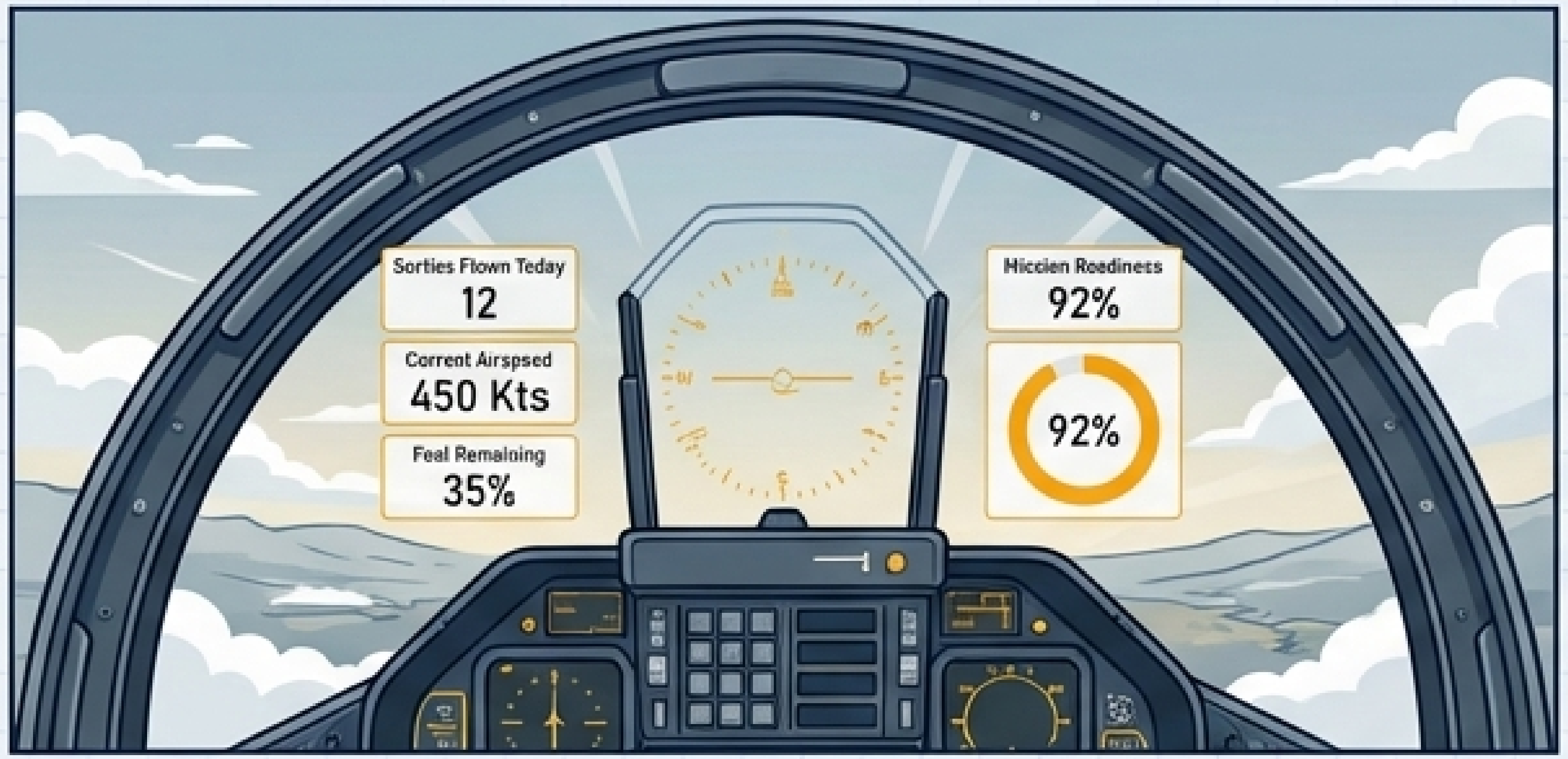
Every transformation you make in the user-friendly interface is recorded as a step in a script using the M formula language.



The Pilot's HUD vs. Post-Flight Analysis (Dashboards vs. Reports)

In Power BI, 'Dashboard' and 'Report' are not interchangeable. They serve two distinct, complementary purposes in the decision-making process.

The Dashboard (The HUD)



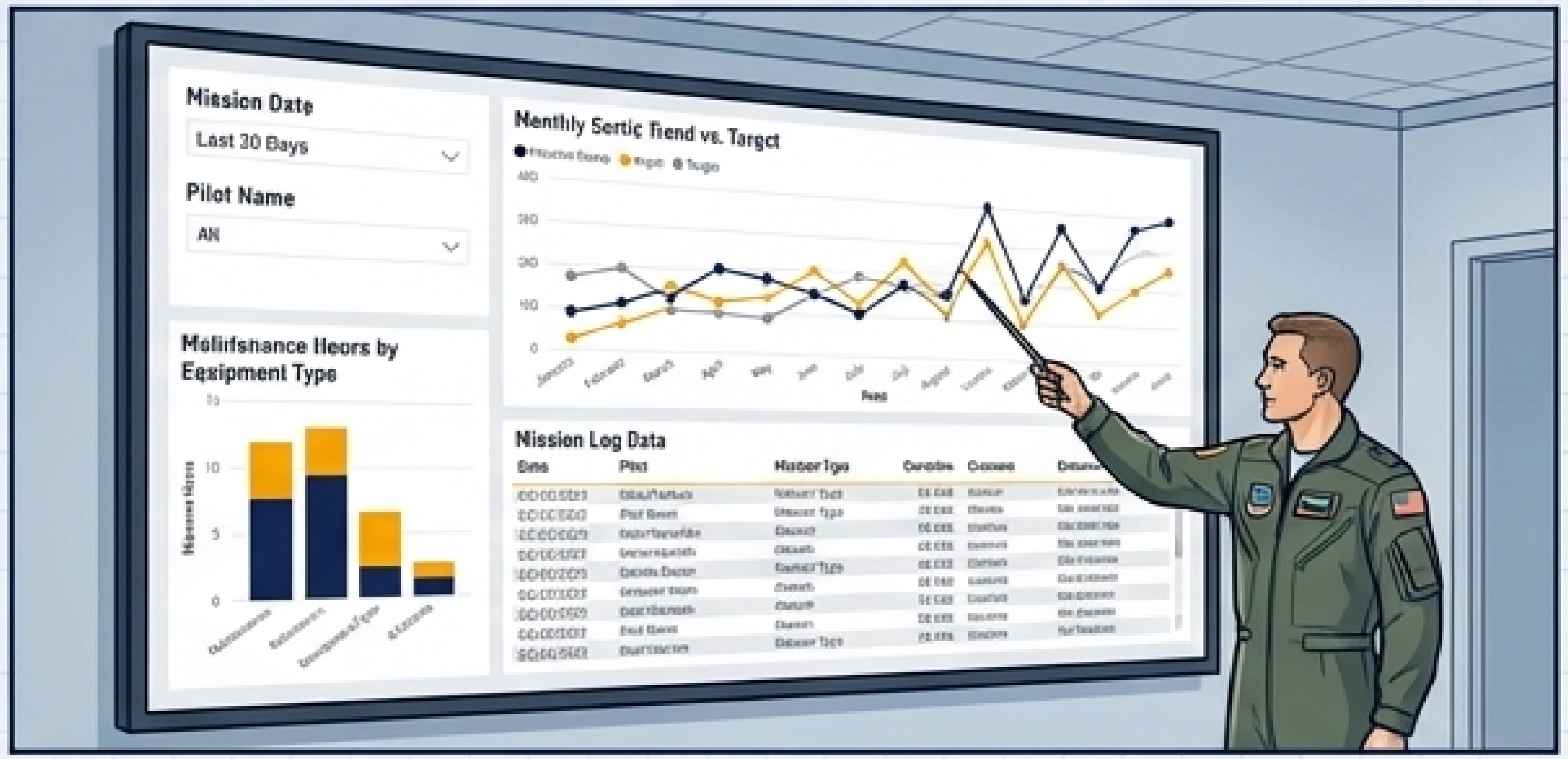
Purpose

At-a-glance monitoring. A single pane view of the most critical, up-to-the-second KPIs.

Characteristics

- Single page (canvas).
- Tells a story on one screen.
- Ideal for tracking real-time metrics.
- Clicking a tile takes you to the underlying report for details.

The Report (The Post-Flight Analysis)



Purpose

Deep-dive analysis and exploration. Allows for comprehensive investigation into *why* something happened.

Characteristics

- Multiple pages.
- Fully interactive with filtering, slicing, and drill-through.
- Directly connected to the full data model for detailed exploration.

Dashboards are built by "pinning" visuals from reports. The dashboard provides the "what," and the report provides the "why."

Your First Flight: Building the Squadron Readiness Dashboard

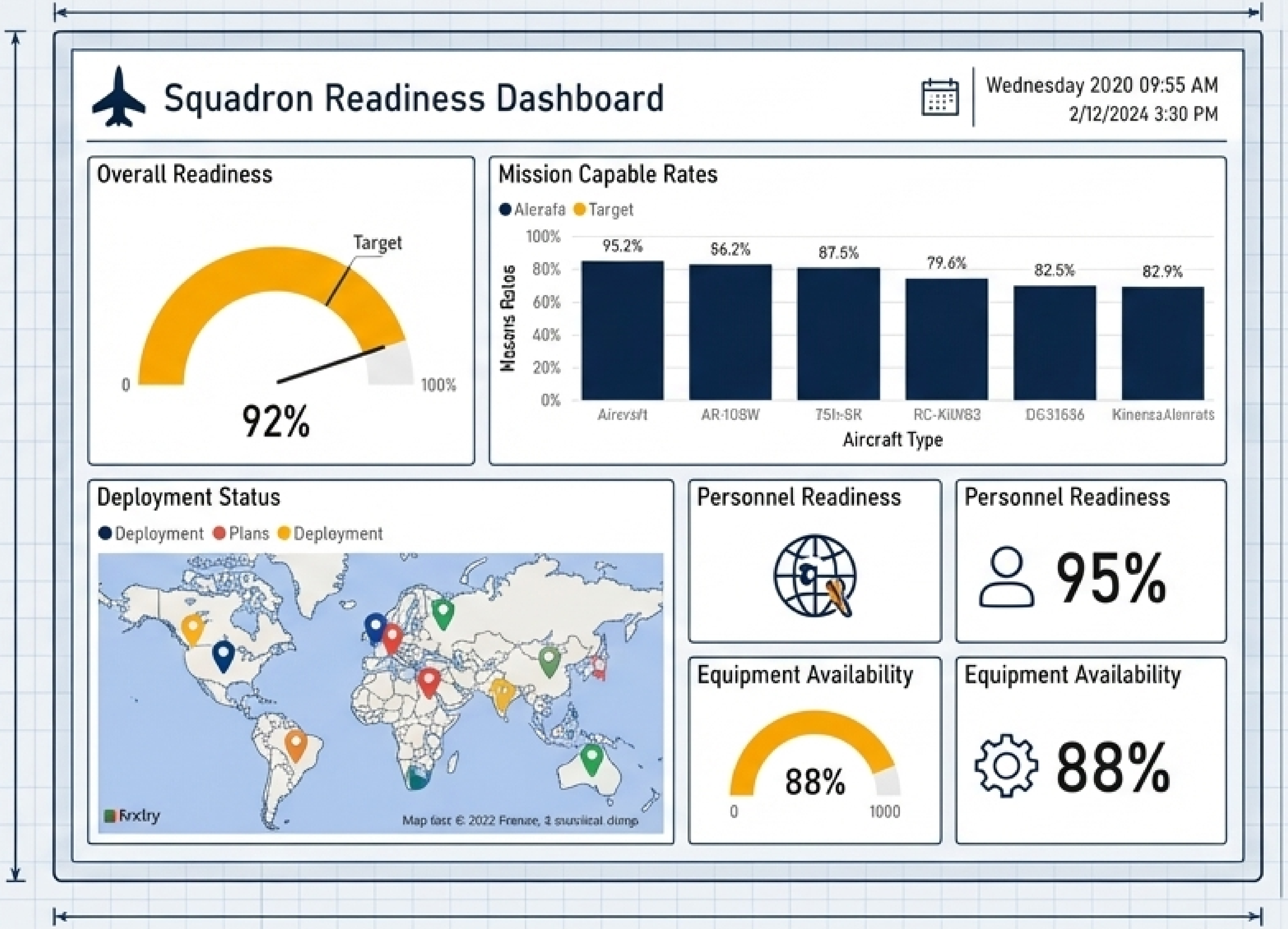
Mission Objective

In the following steps, we will build a basic but effective Squadron Readiness Dashboard. Our goal is to create a single-pane “Heads-Up Display” to monitor key readiness indicators.

Data Sources (Dummy Data)

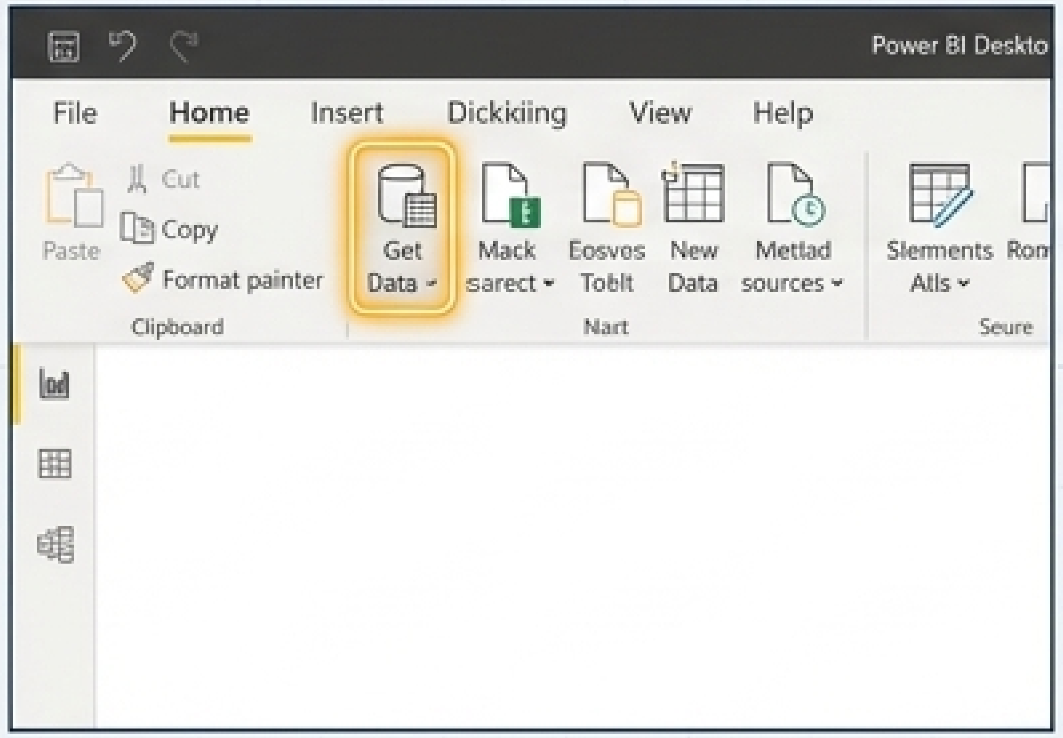
- CSV Personnel_Status.csv
- XLSX Equipment_Maintenance.xlsx
- XLSX Mission_Capable_Rates.xlsx

Open Power BI Desktop. Let’s begin.

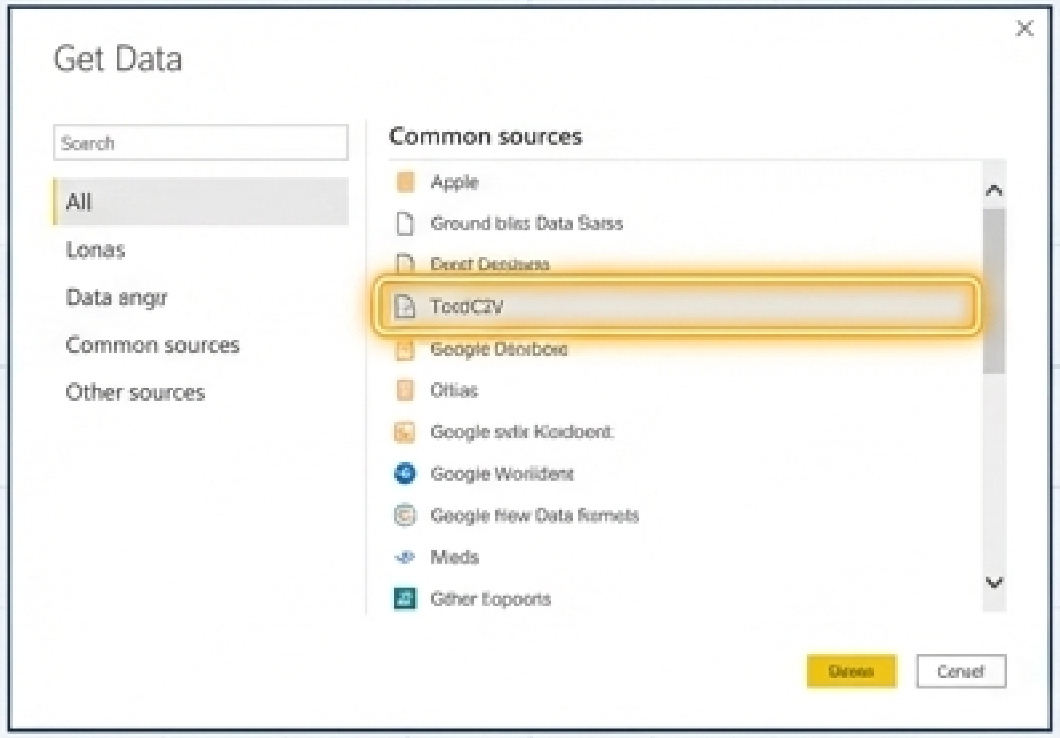


Step 1: Acquire Data Feeds

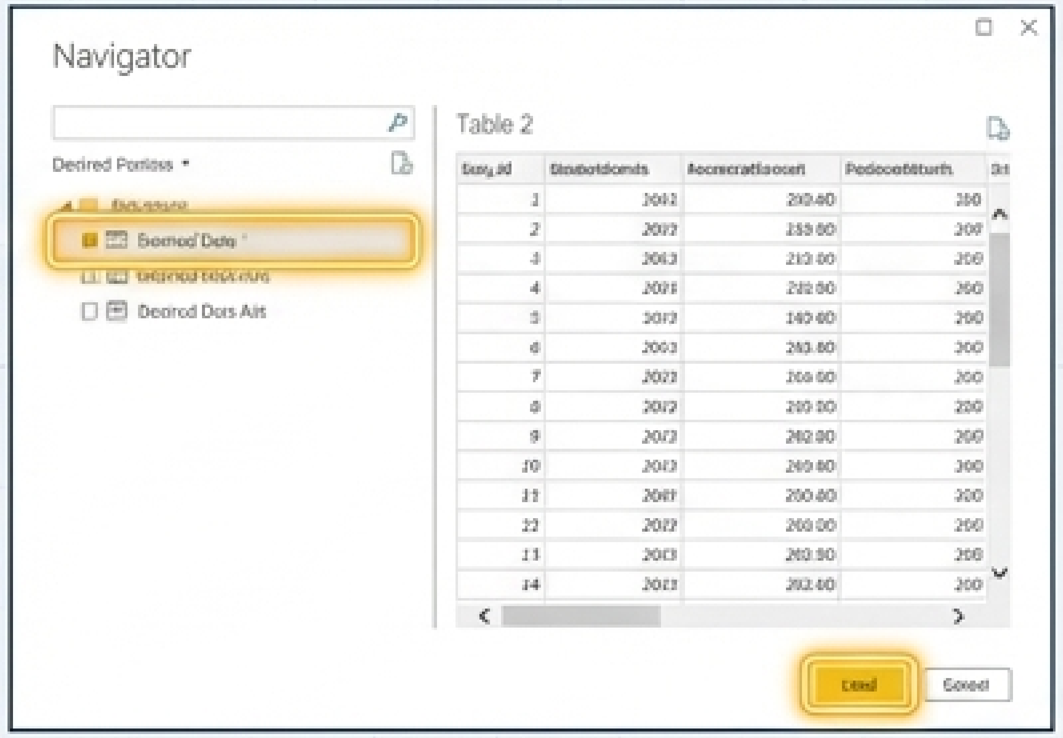
The first step is to connect to our data sources. In Power BI Desktop, the “Home” ribbon provides access to the “Get Data” function.



1. Select ‘Get Data’.



2. Choose your data source type (e.g., ‘Text/CSV’).

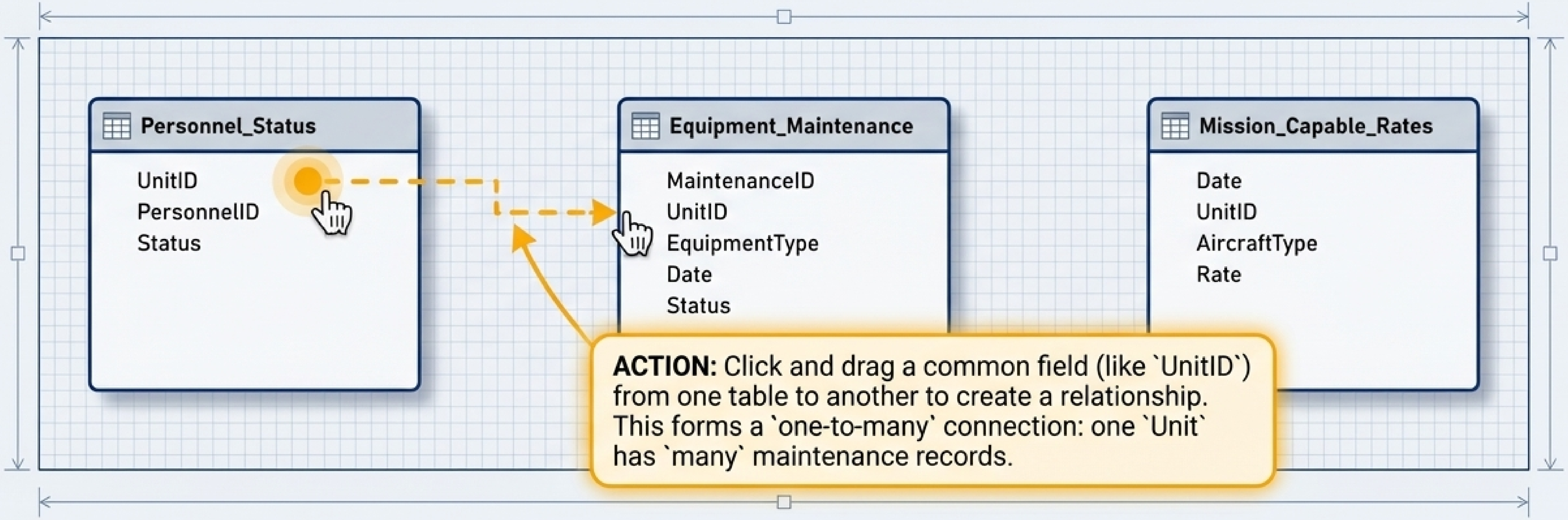


3. Preview and Load the data.

Producer Note: Repeat this process for the two .xlsx files using the ‘Excel workbook’ connector. Once complete, you will see all three tables listed in the ‘Data’ pane on the right side of your screen.

Step 2: Assemble the Blueprint (Model the Data)

With our data loaded, we must connect the tables. This tells Power BI how they relate to each other, forming the "blueprint" for our analysis. We will do this in the "Model" view.



Producer Note: For this exercise, our data is pre-cleaned. In a real-world scenario, you would use the Power Query Editor (your ‘maintenance crew’) to clean and shape the data before this step.

Step 3: Build Your Heads-Up Display (Visualize the Data)

Return to the ‘Report’ view. We will now build the individual components of our dashboard by selecting visuals from the ‘Visualizations’ pane and dragging fields from the ‘Data’ pane.

A. Mission Capable Rate (KPI Card)

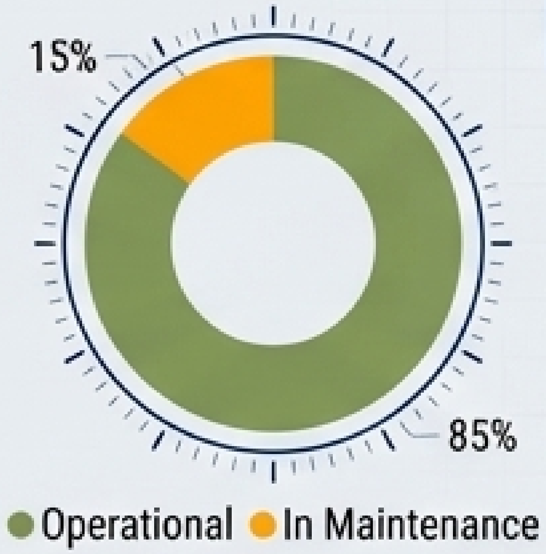
1. Select the ‘Card’ visual.
2. Drag `MC_Rate’ to the ‘Fields’ well.

92.5%

Mission Capable Rate

B. Equipment Status (Donut Chart)

1. Select ‘Donut chart’.
2. Drag `Equipment_Status’ to ‘Legend’ and ‘Equipment_Count’ to ‘Values’.



C. Personnel by Status (Table)

1. Select ‘Table’.
2. Add `Unit’, `Personnel_Name’, and `Status’ to ‘Columns’.

Unit	Personnel_Name	Status
Unit A	John Doe	Active
Unit B	Jane Smith	On Leave
Unit C	Mike Brown	Active

D. Readiness Trend (Line Chart)

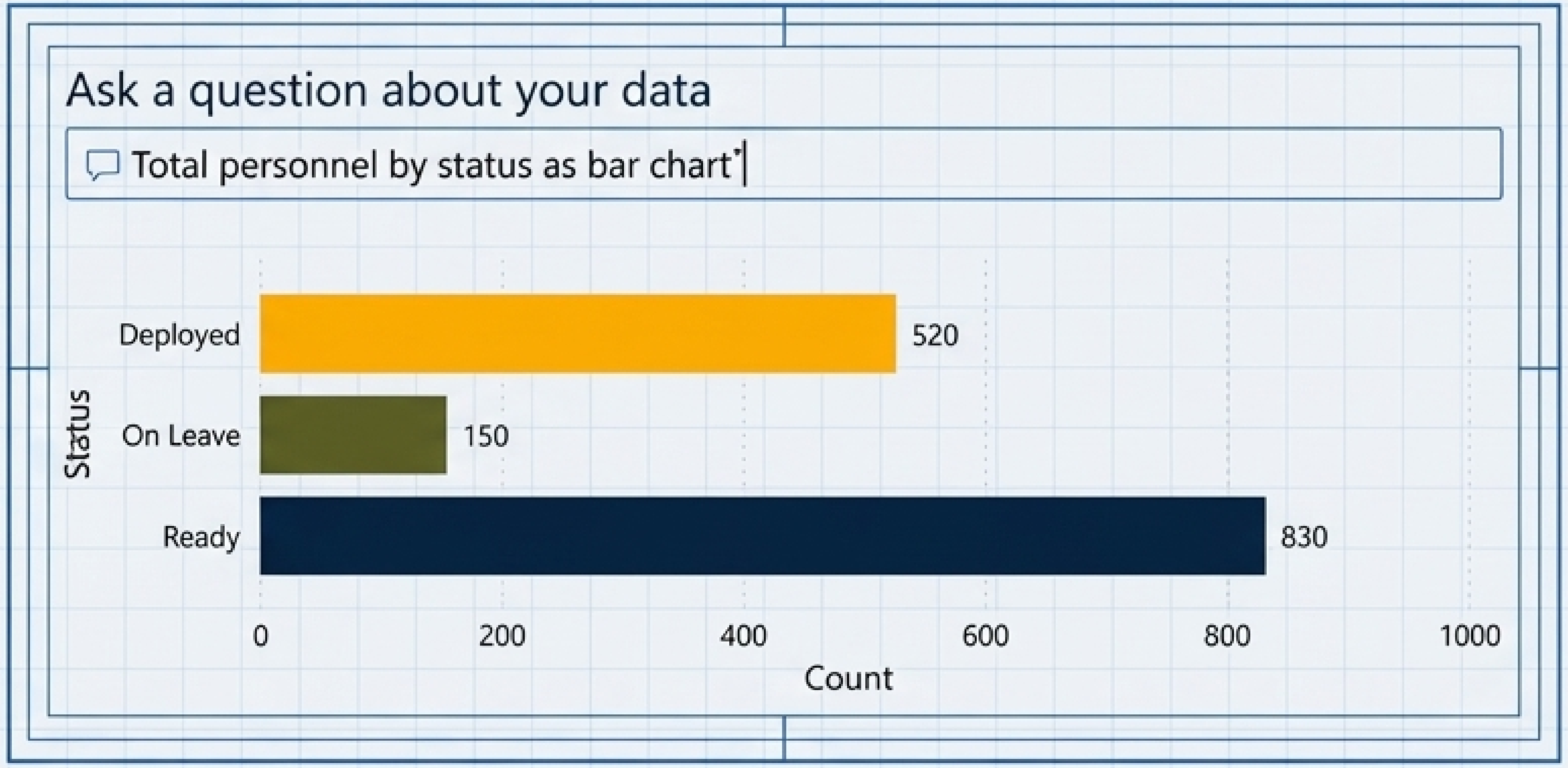
1. Select ‘Line chart’.
2. Drag `Date’ to X-axis and `MC_Rate’ to Y-axis.

Step 4: Enhance with Direct Interrogation (Natural Language Q&A)

Power BI allows you to ask questions of your data in plain English. The Q&A visual can generate insights on the fly without manual configuration.

How to Use It

1. Select the **Q&A visual icon** from the Visualizations pane.
2. In the question box, simply type what you want to know.
3. Power BI will interpret your question and choose the best visual to display the answer.



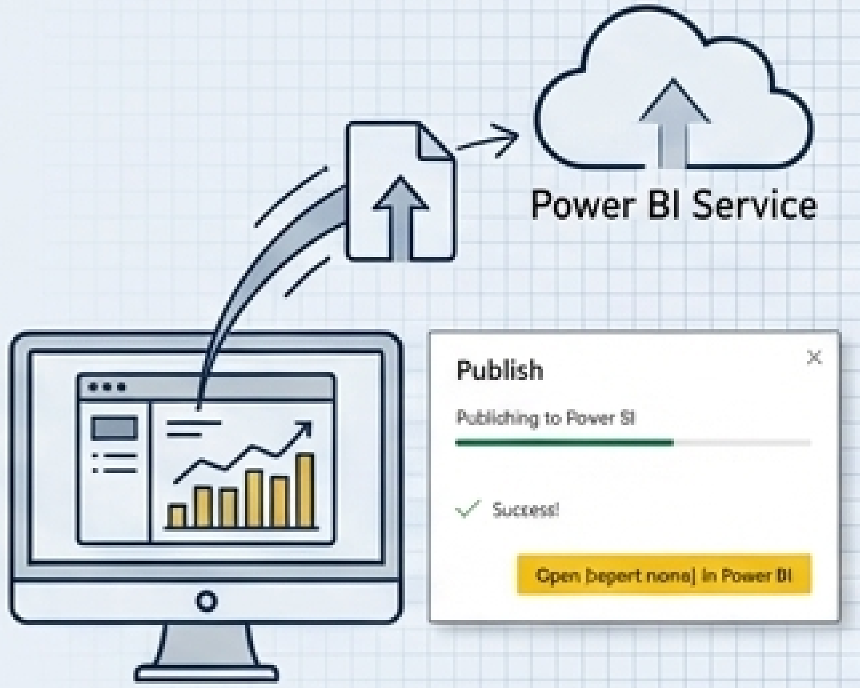
Key Benefit: This is a powerful tool for ad-hoc analysis, allowing commanders to get immediate answers to emergent questions without needing a data analyst.

Step 5: Publish and Automate Your Intelligence

1. Publishing Your Report

Once your report is complete, you publish it to the Power BI Service to share it and create your final dashboard.

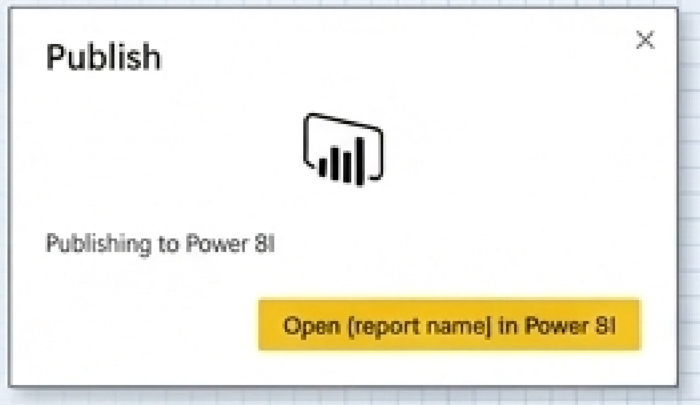
In Power BI Desktop, click the **Publish** button on the 'Home' ribbon. Select your workspace to deploy it.



2. Creating the Dashboard

In the Power BI Service, open your report. Hover over any visual and select the **pin icon** to add it to a new dashboard.

Arrange your pinned tiles too your practice to prevent their dashboard without pinned tiles to create your final single-pane HUD.



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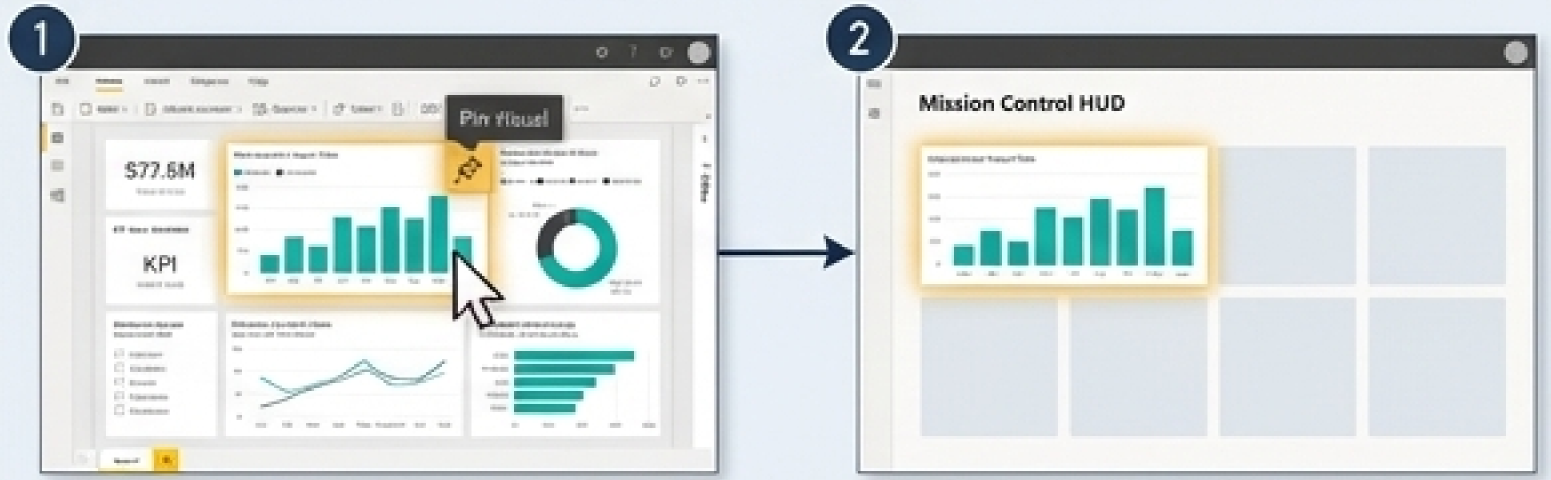
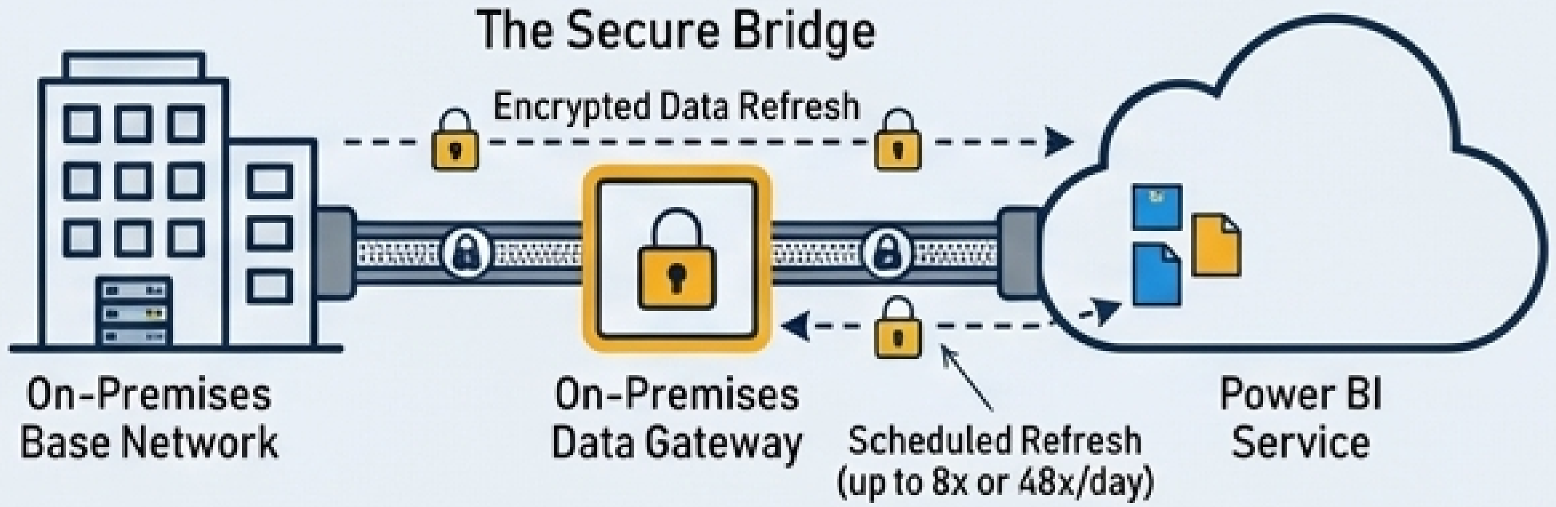


3. Keeping Data Current (Scheduled Refresh)

For data using **Import** mode, you must schedule refreshes to keep it current. Pro licenses allow up to **8** refreshes per day; Premium allows up to **48**.



Mission Complete: You Have Achieved Data Dominance

You have successfully navigated your first flight: from understanding the mission and your equipment to acquiring, modeling, visualizing, and publishing an interactive readiness dashboard. This is the first step in transforming how we leverage data for command decisions.

Your Next Steps & Resources

Continue Your Training

- Microsoft Learn: Follow the "Get started with Microsoft data analytics" learning path for a deeper dive.
- Power BI Community: Join the forums to ask questions and learn from experts.

Practice

Use the provided sample data to experiment with different visuals and DAX measures.

Apply

Identify a data source in your own unit. What is one question a simple dashboard could help you answer? Start small, build confidence.

The ability to see first, understand first, and act first is the ultimate advantage. Use this new capability to build it.